

Can Karakoc

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UC Berkeley Computer Science and Data Science graduate building AI agents, data pipelines, and embedding-based ML systems for biological applications — from protein structure analysis to single-cell transcriptomics. Skilled in Python, SQL, machine learning, statistical modeling, causal inference, and vector retrieval, with a product-minded approach to turning ambiguous data problems into reliable, user-facing systems. Seeking Data Science, Machine Learning, and AI Engineering roles at early-stage biotech and computational biology startups, where I can move fast and own problems end-to-end.

Technical Skills

Programming: Python, SQL, R, Spark

ML/AI: scikit-learn, PyTorch, TensorFlow, Embeddings, Vector Search, FAISS, Neural Networks, PCA, Clustering

Statistics & Modeling: Causal Inference, IPW/SIPW, GLMs, Hypothesis Testing, Cross-Validation, Bootstrap CIs, Model Evaluation

Computational Biology: Single-cell RNA-seq analysis, transcriptomics, protein sequence & structure analysis, omics data processing

Data/Tools: ETL Pipelines, Feature Engineering, PostgreSQL, NoSQL, Caching, Tableau, Power BI, Plotly, Git, Linux, Jupyter

Experience

AI Engineering Intern – Cendra | April 2026 – Present | Remote

- Shipped an autonomous AI voice agent for outbound sales, qualifying intent, updating CRM state in real time, scheduling callbacks, and booking demos; projected to replace 3+ SDR workflows and reduce CAC by ~10x.
- Built agent workflows around enriched lead data, missing-context recovery, post-call classification, and human escalation logic, driving sprint-level product decisions in a fast-moving startup environment.

Website Content & Design Manager – UC Berkeley Development Engineering | June 2024 – Aug 2025 | Berkeley, CA, USA

- Managed 20+ production web pages for Berkeley's Master of Development Engineering program, improving accessibility, information discoverability, and stakeholder publishing workflows.
- Built scalable CMS structures and reusable content components, using analytics and user behavior signals to reduce manual updates and translate stakeholder requirements into maintainable web updates.

Data Analysis Intern – Velocity Inc. | May 2023 – Aug 2023 | Istanbul, Turkey

- Built Python pipelines processing millions of mobile location records for behavioral segmentation, cohort analysis, and product strategy use cases.
- Engineered 20+ temporal and mobility features from raw event-level data; automated weekly analytics workflows and Tableau dashboards used by product and strategy teams.

Projects

Protein Interaction Visualizer & Analysis Platform | Personal Project | [App](#) | [GitHub](#)

- Built a full-stack structural biology web app (Next.js/FastAPI) computing 18+ protein structure metrics — contacts, interfaces, binding pockets, and antibody CDRs — for AI-predicted and experimental structures.
- Designed a privacy-first, stateless backend with client-side IndexedDB caching, enabling self-hosted deployment for sensitive structures with no cloud storage or user accounts.
- Implemented custom computational geometry (VDW clash detection, SASA, pocket detection) with NumPy/SciPy, Gemmi, and RDKit, backed by 40+ pytest tests; deployed on Vercel + Render.

Protein Embedding-Based Similarity Search System | Personal Project | [App](#) | [GitHub](#)

- Built a vector retrieval system that embeds protein sequences into dense representations and retrieves nearest-neighbor matches with FAISS similarity search — a modular pipeline spanning sequence validation, embedding generation, caching, indexing, and ranked retrieval, applicable to target identification and homology search.
- Implemented cache-hit/miss diagnostics and tests to reduce redundant computation and validate retrieval behavior across repeated searches.

Single-Cell RNA-seq Analysis | Personal Project | [GitHub](#)

- Processed high-dimensional scRNA-seq data with HVG selection, regression/scaling, PCA/UMAP, clustering, and marker-based interpretation.
- Trained classifiers for immune cell-type prediction, reaching 95% accuracy and producing interpretable diagnostics for model performance and biological signal.

Energy Production and Carbon Emissions | DATA 102, UC Berkeley | [GitHub](#) | [Paper](#)

- Built predictive models for state-level renewable energy production using GLMs, Random Forest, and KNN with cross-validation-based model comparison.
- Developed a causal inference pipeline estimating the effect of electricity prices on carbon emissions using IPW/SIPW, propensity modeling, and bootstrap confidence intervals.

Education

University of California, Berkeley | Berkeley, CA

B.A. Computer Science; B.A. Data Science | Aug 2021 – May 2025

GPA: 3.8/4.0

Relevant Courses: Designing, Visualizing, and Understanding Deep Neural Networks; Introduction to Machine Learning; Data, Inference & Decisions; Data Engineering; Optimization Models in Engineering; Introduction to Database Systems; Machine Structures; Data Structures.

Leadership

Teaching Assistant – Human-centered Design DeCal | May 2023 – Aug 2023 – Led weekly sessions for 60+ students on design methodology, research synthesis, and prototyping.

Design Consultant – Berkeley Innovation | Jan 2023 – May 2024 – Delivered user-centered design solutions for clients through structured research and iteration.